

HANYANG UNIVERSITY · DEPT. OF NUCLEAR ENGINEERING · THINK LAB · NUE4067

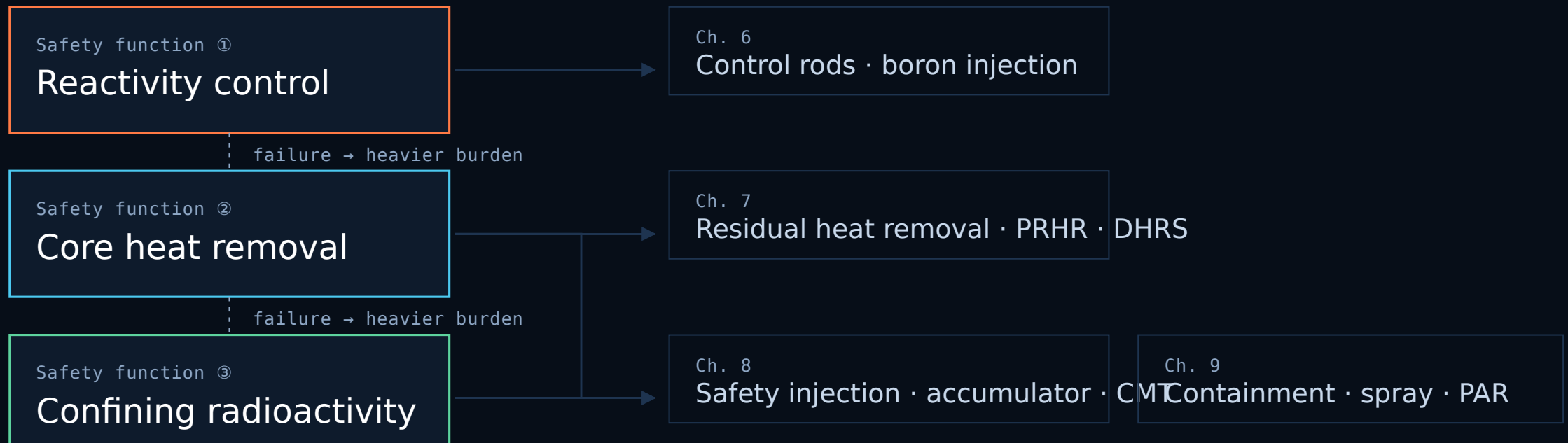
SMR **Design Requirements** and Engineered Safety Features

The minimum set of principles needed to answer with numbers why the design looks the way it does

Weeks 2-3 · Instructor Minseop Song · Rev1.0 · September 2026

The **three things** a reactor must do - and they call for the systems

- Memorising system names first tells you nothing about why they exist. Read it as **requirement** → **function** → **system**
- It is a **chain**: when an upstream function fails the burden on the next one grows - the order carries meaning

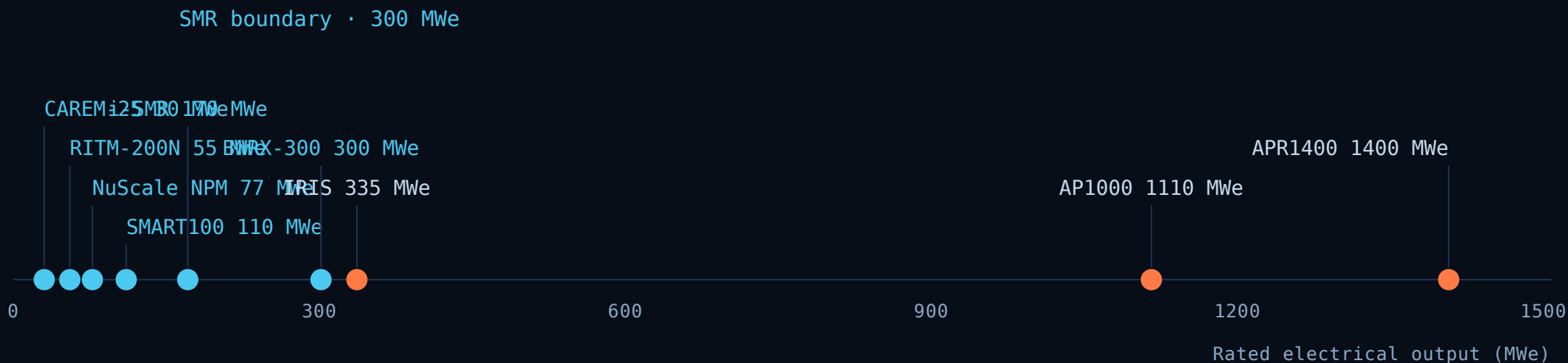


01 What is an SMR

Two words in the name, and the price paid for the choice to build it small.
Miss this chapter and you miss the motive behind every design choice that follows.

Small is a choice, not an outcome

- **Small** - 300 MWe or less. Note the yardstick is **electrical** output, not thermal
- **Modular** - ① a **fabrication unit** built in a factory and shipped, ② a **power unit** added one at a time
- Factory fabrication demands it fit inside transport limits, which demands it be small - **small is what makes modular possible**



Shrink the output and **cost per kW rises** - how much it rises is the whole question

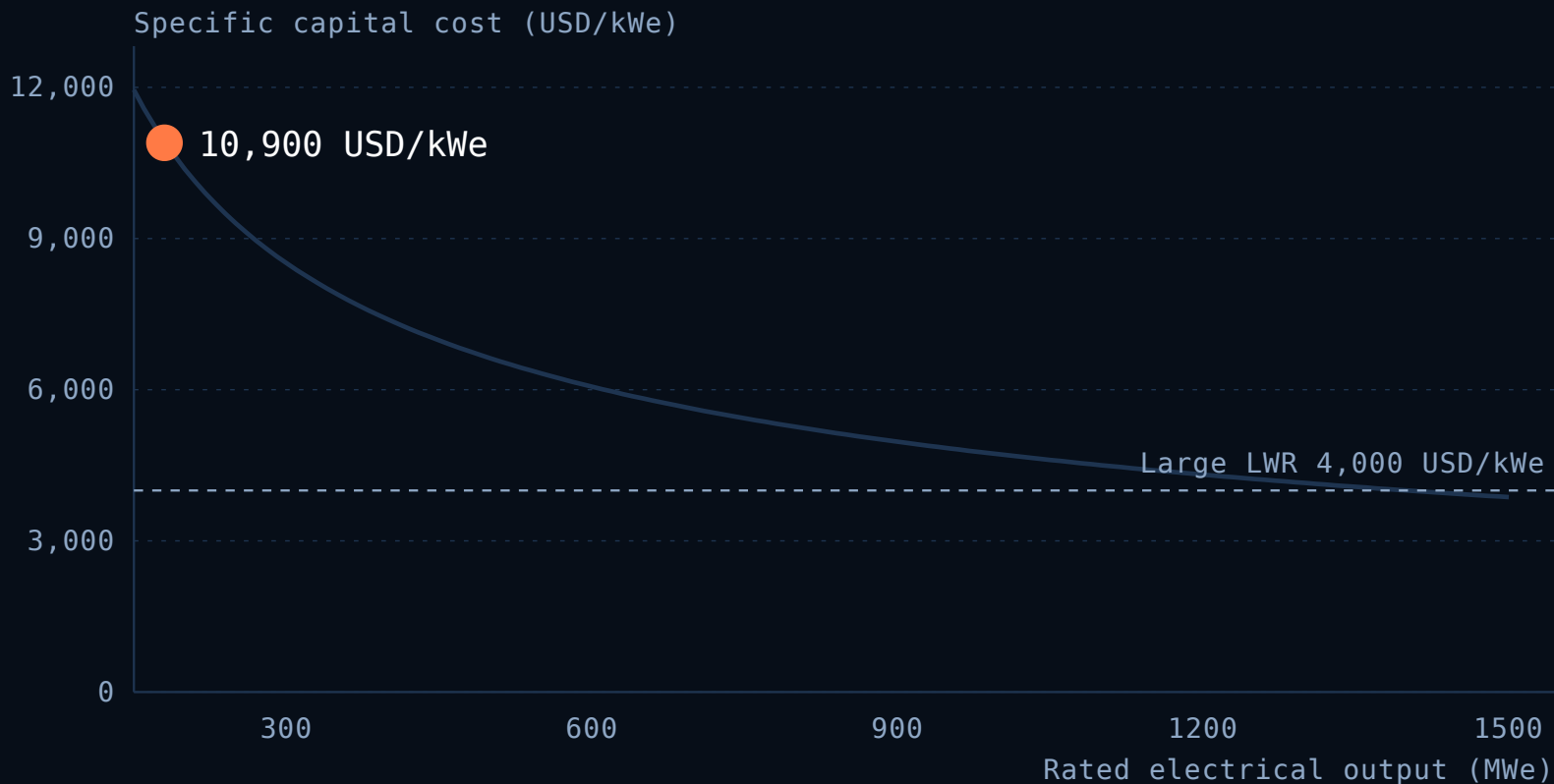
- Vessel, piping and structure cost follows **surface area** while power follows **volume** → scaling exponent $n \approx 0.4-0.7$
- Watch **cost per kW**, not total capital cost. The two move in opposite directions

$$C_2/C_1 = (P_2/P_1)^n \rightarrow (C/P)_2/(C/P)_1 = (P_2/P_1)^{n-1}, \quad n-1 < 0$$



Four factors that claw back the 2.7x - press a button to apply them one by one

- Every one of them assumes **repetition**. Nowhere in the world is an SMR being built repeatedly yet - all of them are **FOAK**



180 MWe · CURRENT ESTIMATE

10,900 USD/kWe

① Co-siting · multiple units

② Modular design · factory build and learning

③ Plant configuration · site layout

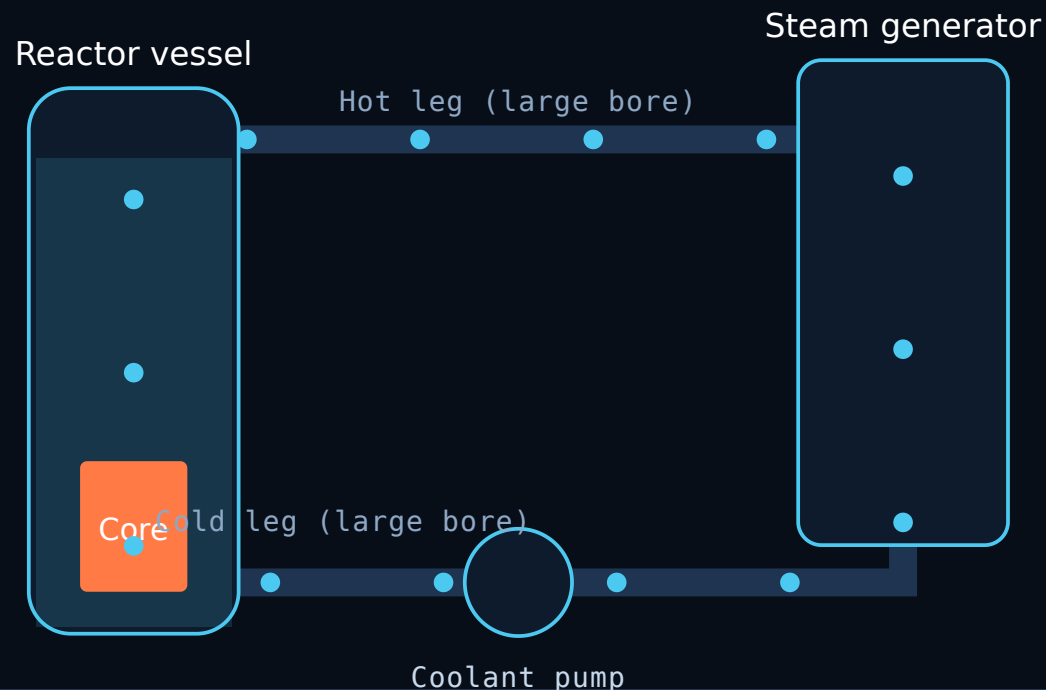
④ NOAK · mature stage

Reset all

Shared indivisible cost · 15 % on unit 1 plus 5 % each on units 2-4 · learning factor 0.81-0.90 for a four-unit site · 4.5 % per doubling of rating

Loop type - a single large pipe defines the worst design basis accident

- A **large-bore hot leg and cold leg** join the vessel to the steam generator → break one and you have a **large LOCA**
- The coolant is driven by **pumps**. Stop the pumps and the flow is gone at once



CIRCULATION DRIVEN BY

Pumps

LEVEL ABOVE CORE

100%

Break the hot leg (LBLOCA)

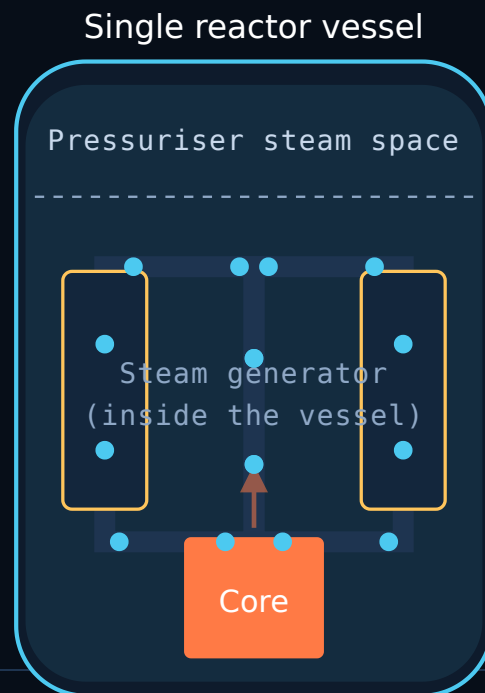
Back to normal operation

Once it breaks, depressurisation, level drop and loss of heat transfer proceed at the same time. Quantified in Ch. 8.

Schematic drawn for this course. Pipe count and equipment layout are simplified and the scale is arbitrary. GDC 4 allows dynamic effects to be excluded where break probability is extremely low (the basis for LBB)

Integral type - the accident is **designed out** rather than mitigated

- Core, steam generator and pressuriser all sit in **one vessel** → no large-bore pipe exists
- Circulation without pumps, driven by **density difference**. The driving head is about 1/50 of a large PWR pump (calculated in Ch. 2)



CIRCULATION DRIVEN BY

Density difference

LARGE-PIPE BREAK

Designed out

Try breaking it

There is a price: a large forged vessel, more penetrations, and poor maintenance access to the in-vessel steam generator.

Reading the sales pitch as an argument

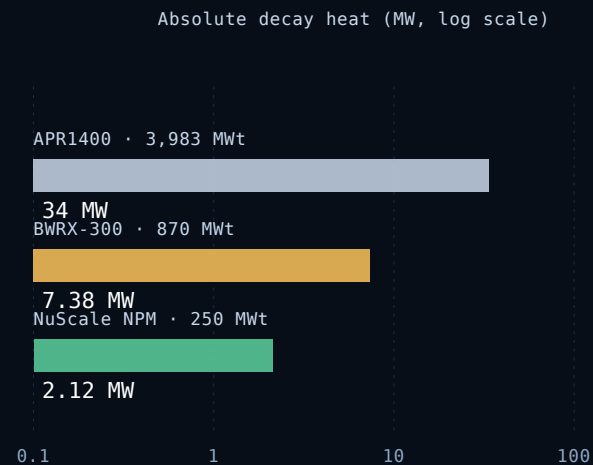
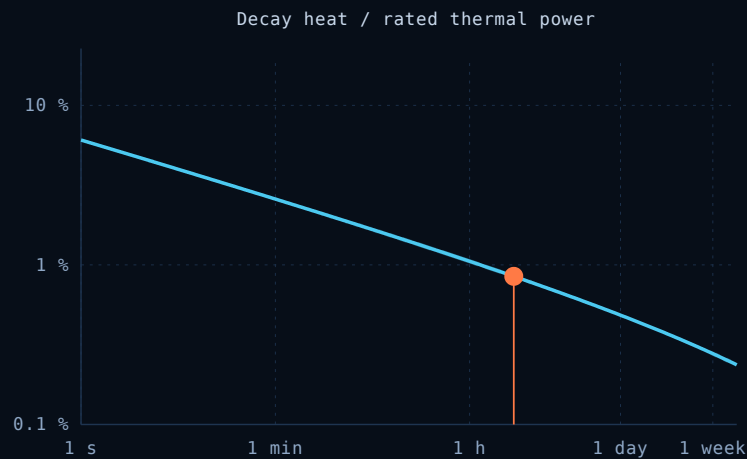
COMMON CLAIM	WHAT ACTUALLY SUPPORTS IT	WHAT TO WATCH
Small, so easy to ship and install	Size fits inside transport limits → factory fabrication becomes possible	A shippable size is not automatically an economic size
Easy to scale up or down	Add modules to add output and follow demand growth	Licensing and operator staffing for many modules at once is a separate problem
Safety improves	Less decay heat in absolute terms, natural circulation becomes viable, large pipes are eliminated	Not “safer” but “safe in a different way” - quantified in Ch. 2
Capital cost is lower	Total capital cost in absolute terms is lower → less investment risk	Cost per kW is in fact higher. Always keep the two apart
Wider range of applications	Heat and power together; remote sites, industrial parks, ships	Most of it is still short of demonstration

02 What changes when it gets small

Two pieces of physics carry the whole SMR safety argument: the absolute amount of decay heat, and the driving head of natural circulation. These two numbers set every system design that follows.

The fraction is the same. The **absolute amount** is not - run the slider and watch

- The decay heat **fraction** barely depends on core size. One hour after shutdown any reactor sits near 1 % of rating
- What differs is the **absolute load** - 1 % of 3,983 MWt is 40 MW, 1 % of 250 MWt is 2.6 MW. A factor of **15**



TIME SINCE SHUTDOWN

2.5 h

FRACTION OF RATING

0.85 %

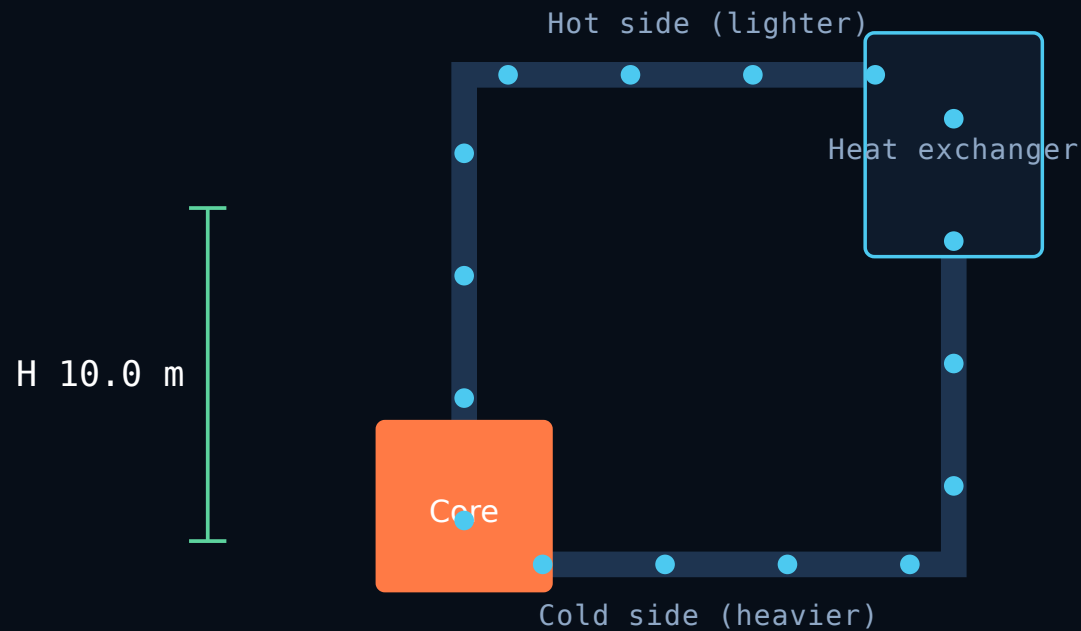
TIME

2.5 h

1 s to 1 week. The scale is logarithmic, so one step of the slider covers more time the further right it goes.

With no pump, **density difference** has to push - so how hard does it push

- The only driving force is buoyancy from the thermal-centre height H and the temperature difference ΔT
- A small driving head means **small flow** → to remove the same power the **core temperature rise must grow**



Driving pressure, kPa

Large PWR pump 600

Nat. circ. 10.8 kPa

DRIVING HEAD ΔP

10.8 kPa

CIRCULATION VELOCITY

1.07 m/s

CORE ΔT · 250 MWT

29 K

THERMAL-CENTRE HEIGHT

10.0 m

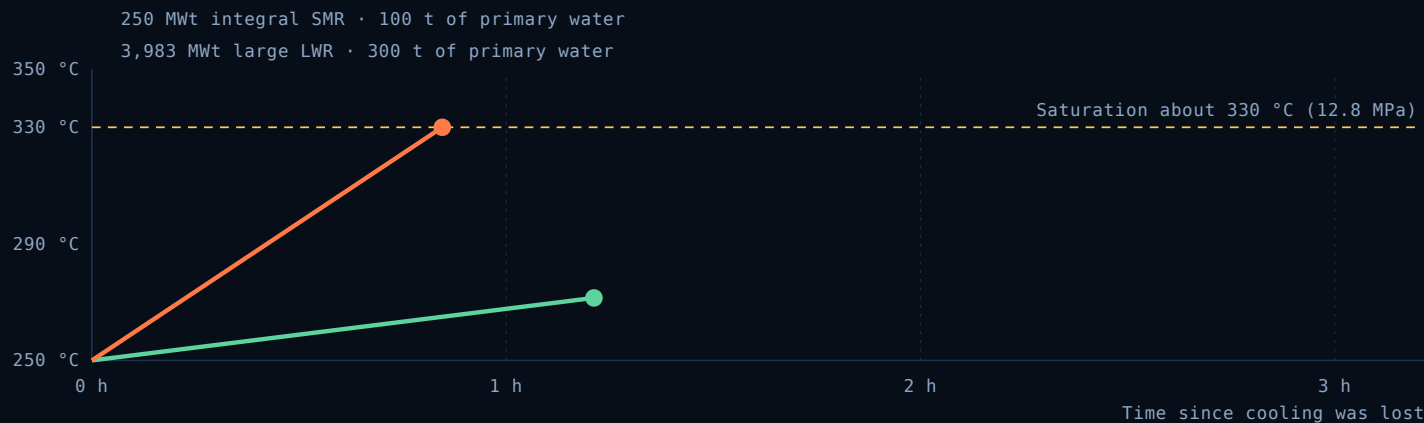
CORE INLET-OUTLET ΔT

67 K

Calculation $\Delta p = (\rho_c - \rho_h) g H$, $u = \sqrt{[2\Delta p / \rho(fL/D + \Sigma K)]}$ · compressed-water density fit (10-16 MPa) · loop $K = 25$, flow area 2 m^2 , cold side at 250°C . The 600 kPa pump head is a representative figure for a large PWR

What “more time to act” physically means

- Less decay heat and more water means the temperature climbs **slowly** - and that slowness is the time operators and passive systems get to use
- SMR 17.7 K/h against large LWR 94.6 K/h - **about five times**



SMR TEMPERATURE

271 °C

LARGE LWR TEMPERATURE

330 °C

ELAPSED TIME

1.2 h

The gap between when the two curves reach the same saturation line is the difference in time available to act.

Calculation $dT/dt = P_d / (M c_p)$, $c_p = 5.3 \text{ kJ}/(\text{kg}\cdot\text{K})$, using decay heat one hour after shutdown (2.6 MW for the SMR, 41.8 MW for the large unit). This is the most conservative bound, with cooling completely lost; in reality vessel heat loss and the residual heat removal system also act

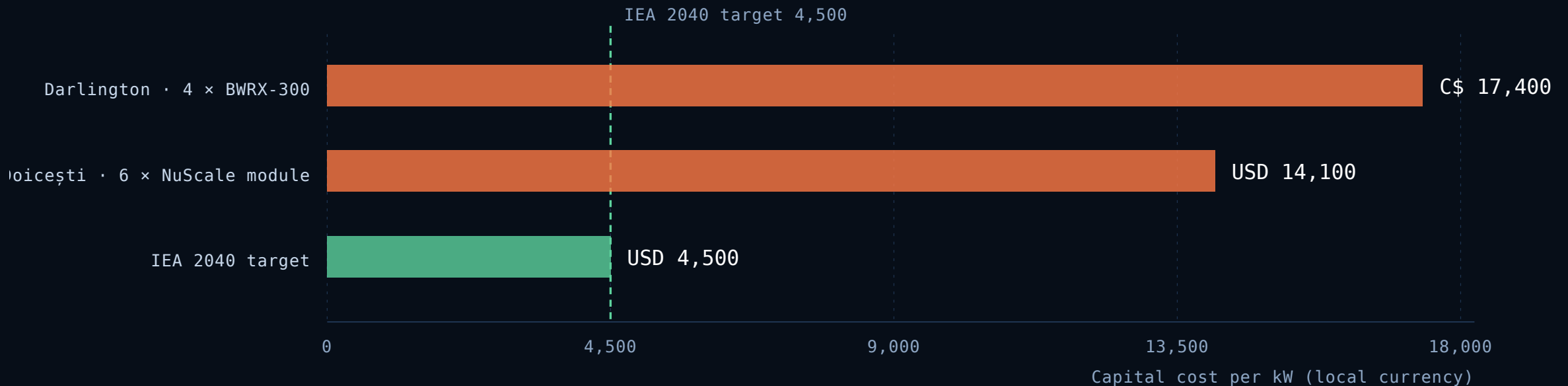
03

Open issues

Economics, regulation, safety, feasibility - what has been settled and what has not. Every status in this chapter is as of September 2026; state it without that date and it will be wrong in six months.

Every reduction factor assumes **repetition** - and so far there is **only FOAK**

- Darlington, four BWRX-300 units, C\$20.9 bn · about 1,200 MWe → **about C\$17,400 per kW**
- Doicești, six NuScale modules, USD 6-7 bn · 462 MWe → **about USD 13,000-15,200 per kW**
- The IEA target for 2040 in the US and Europe is USD 4,500/kW with an LCOE of 60-80 USD/MWh



Source WNISR, "Ontario's Darlington SMR project to cost nearly \$21 billion" (2025) · World Nuclear News, "Final investment decision taken for Romania's SMRs" (2026) · IEA, *The Path to a New Era for Nuclear Energy* (2025). Caution the projects differ in scope (transmission, site, financing) and in currency, so the figures are not directly comparable

The existing rules were written around large light water reactors

- Containment size, emergency planning zone radius, operator staffing, treatment of multiple units - all sized for large plants
- In the Korean i-SMR review, **the module design and the safety of the passive safety systems** are named as focal points

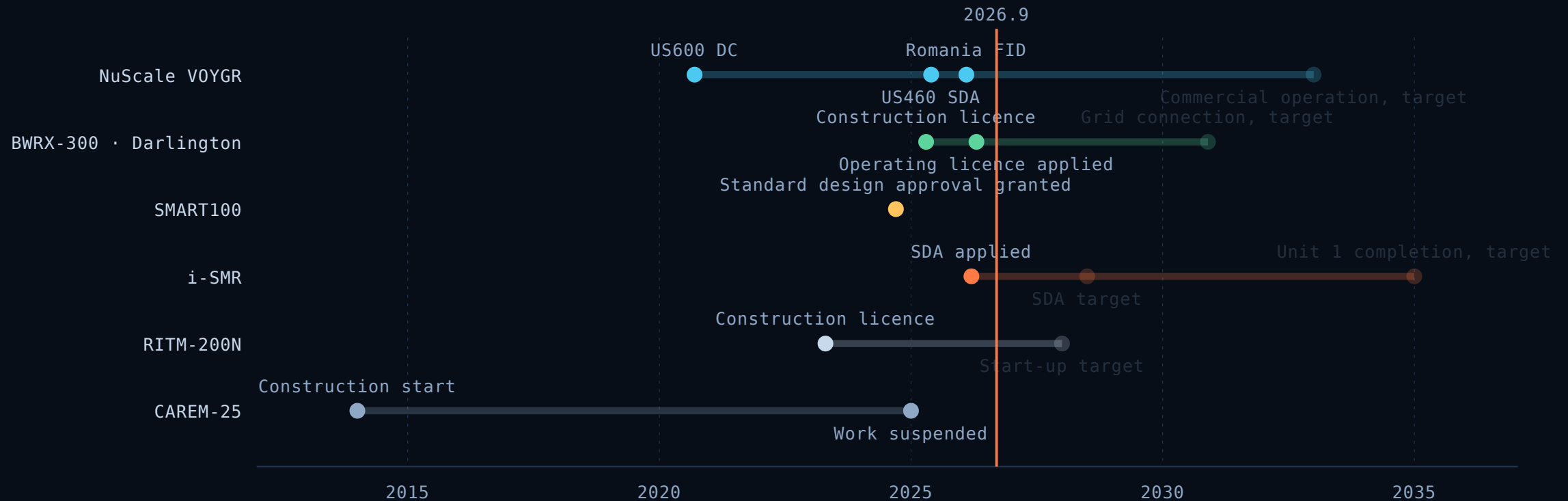
ACTIVITY	STARTED	STATUS (SEPTEMBER 2026)
SMR Regulators’ Forum	2015	Phase 1 graded approach, defence in depth, EPZ → Phase 2 licensing and passive design features → Phase 3 five reports → Phase 4 under way
IAEA NHSI	2022	A regulatory track (less duplicated review) plus an industry track (standardisation) · Phase II implementation
NSSC roadmap for SMR regulation	2026.2	Aims for a comprehensive SMR licensing framework by 2030
i-SMR standard design approval (SDA)	2026.2	Application filed → KINS review plan reported 2026.8 · 24 months by statute · approval targeted for 2028
SMART100 SDA	2019.12	Granted on 26 September 2024

It is not that regulation is “missing”; it is that **the large-plant rules exist and the SMR rules are being written**. Rather than waiting for a rule that is not there, the designer has to argue **why this design meets the requirements that do exist**.

→ Ch. 4 requirements, Ch. 5 ESF design features

Where things actually stand - the red line is now

- Closest to generating power is **BWRX-300** - the first G7 construction licence in April 2025, operating licence applied for and civil works started in April 2026
- CAREM-25 stalled not on technology but on **continuity of funding and organisation**

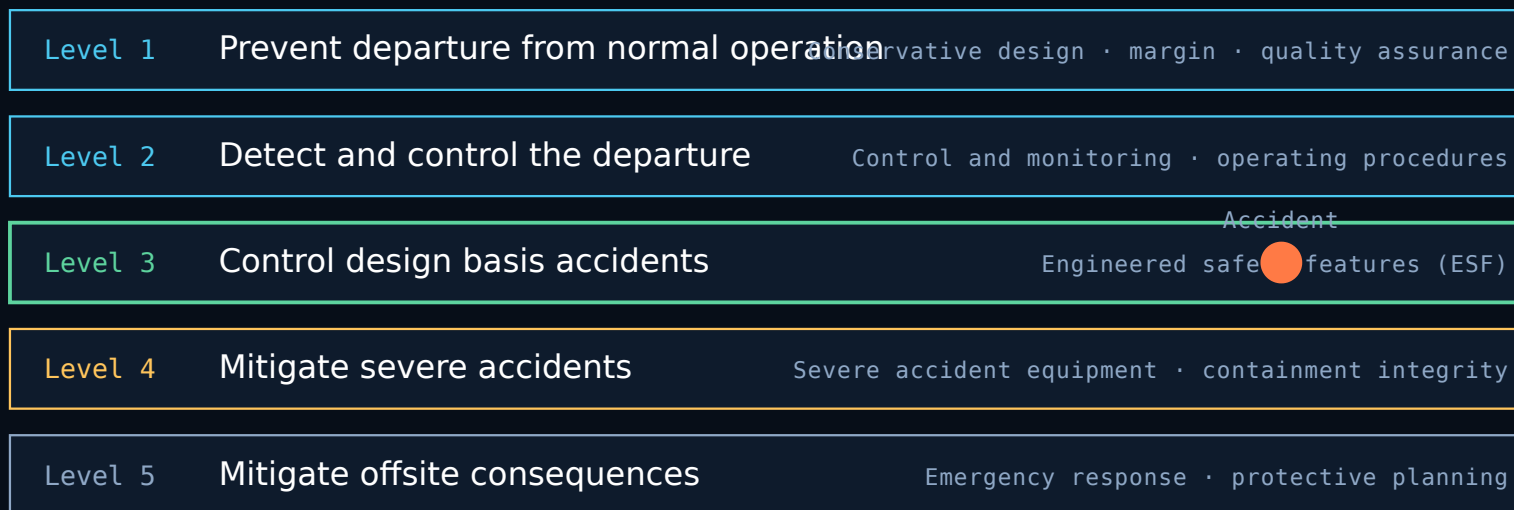


04 The frame of safety requirements

Before looking at individual systems, set up the frame that says why those systems exist at all. Fundamental safety functions, defence in depth, the single failure criterion - these three come back in every chapter that follows.

An accident pierces the **layers** one at a time - where do you make it stop

- Engineered safety features hold **Level 3**. That is the scope of this text
- Splitting into levels means nothing if **one cause kills them together** - which is what happened at Fukushima



Level 3 - the engineered safety features hold the accident inside the design basis

Sound design - replay

Fukushima - common cause replay

One event, an earthquake and a tsunami, disabled offsite power (support for Levels 1-2), the emergency diesel generators (Level 3) and the ultimate heat sink (Levels 3-4) **at the same time**.

→ hence the SMR emphasis on systems that need no electrical power

Design as if one thing **has already failed** - but get three points right

“A single failure means an occurrence which results in the loss of capability of a component to perform its intended safety functions. **Multiple failures resulting from a single occurrence are considered to be a single failure.** Fluid and electric systems are considered to be designed against an assumed single failure if neither (1) a single failure of any **active** component (assuming passive components function properly) nor (2) a single failure of a **passive** component (assuming active components function properly), results in a loss of the capability of the system to perform its safety functions.”

10 CFR 50 Appendix A · Definitions and Explanations

TRAP ①

The single failure criterion sits not in the GDC themselves but in the **definitions** at the head of the appendix. This is where citations most often go wrong.

TRAP ②

An active failure and a passive failure are assumed **one at a time**. Never both at once.

TRAP ③

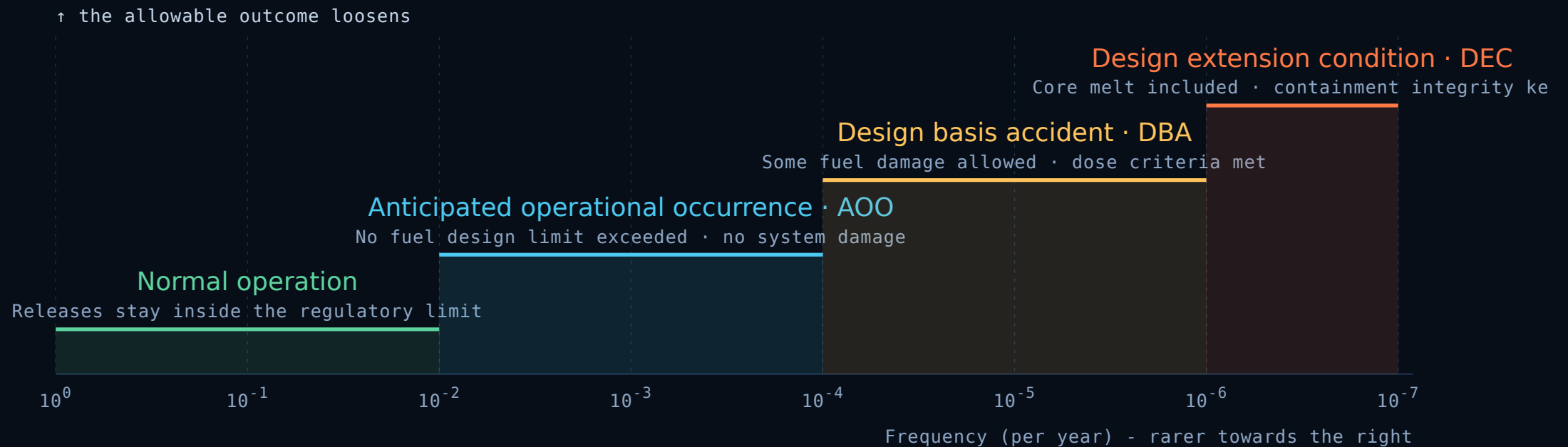
Multiple failures from one occurrence count as a single failure. One dead power panel stopping several valves is still one failure.

STILL OPEN

How the criterion applies to passive components in fluid systems is still marked **under development** in the text. This is where an SMR leaning on passive systems raises a genuinely new regulatory question.

What you design against - frequency sets the allowable outcome

- A transient that can happen every year must not damage fuel; for a once-in-a-million-year event it is enough to **keep containment intact**
- The levels of defence in depth from the previous slide map exactly onto this **frequency-consequence axis**



When stuck, reading the GDC text itself is the fastest way out

GROUP	GDC	TITLE	WHAT IT DEMANDS
Design principles	2	Design Bases for Protection Against Natural Phenomena	The most severe natural phenomena in the site history, plus margin, plus combined effects
	4	Environmental and Dynamic Effects Design Bases	Withstand break reaction, missiles and jets. Dynamic effects may be excluded where break probability is extremely low
	17	Electric Power Systems	Two physically independent offsite circuits plus onsite power
	23	Protection System Failure Modes	On a broken wire or loss of power, fail into the safe state
Reactivity	26	Reactivity Control System Redundancy and Capability	Two independent systems on different design principles · stuck rod assumed
	27	Combined Reactivity Control Systems Capability	Combined with ECCS boron, bring to and hold a coolable shutdown state
Heat removal	34	Residual Heat Removal	Assume a single failure in each case: onsite power only, and offsite power only
	35	Emergency Core Cooling	Adequate core cooling during a LOCA and a limited metal-water reaction
Containment	16	Containment Design	Provide an “essentially leak-tight barrier”
	38	Containment Heat Removal	Bring pressure and temperature down rapidly after an accident and hold them

Source 10 CFR Part 50, Appendix A (verified against eCFR and Cornell LII). The Korean counterpart is the Rule on Technical Standards for Nuclear Reactor Facilities, Articles 20, 23, 24, 29 and 30

55, 56, 57 Containment Isolation Valves One or four isolation valve arrangements on each penetrating line (Ch. 9)

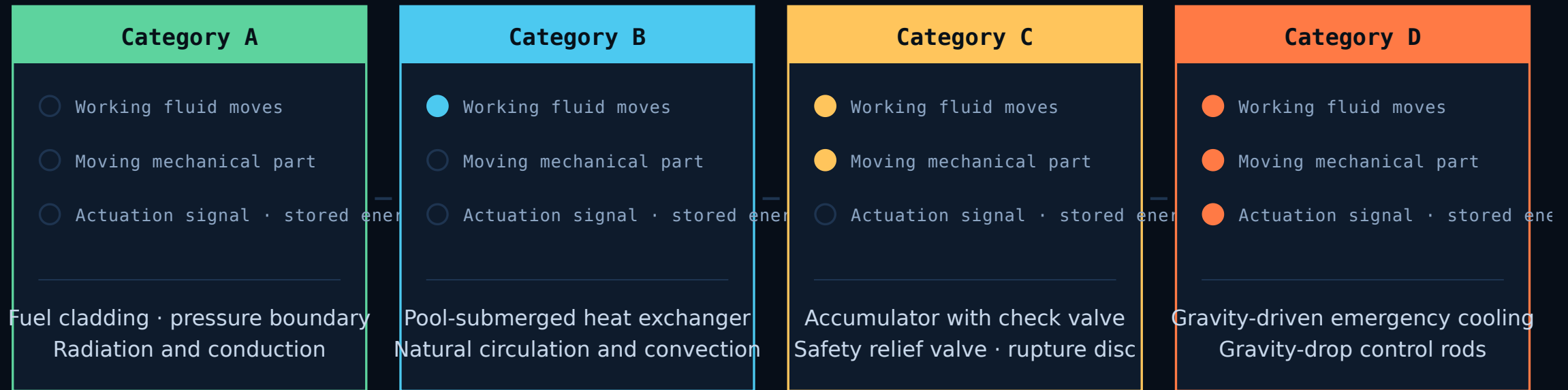
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05 **Engineered safety features**

What an ESF is, how active and passive are told apart, and the five properties a design has to have. This is the shared vocabulary for reading the systems in Chapters 6 to 9.

“Passive” is not a yes-or-no - it splits into four grades

- Moving right through the categories adds **one more thing that can fail** at each step
- Most real SMR “passive safety systems” are **Category C or D** - passive does not mean it cannot fail



← fully passive · moving parts and actuation signals accumulate →

Each of the five properties has a rule behind it

- Learn them together with the rule and they become useful in practice - “why is it built this way” is “which rule demands it”

PROPERTY	WHAT IT MEANS	MATCHING REQUIREMENT
Redundancy	Each train on its own has enough capacity to meet the required function	Single failure criterion · GDC 34·35·38
Diversity	Different principles, vendors or technologies, to escape common cause failure	GDC 22 · GDC 26 · SSR-2/1 Req.24
Independence	Redundant trains are physically separated so one cannot affect the other	GDC 22·24 · SSR-2/1 Req.21
Fail-safe design	Arrange systematically so that failure lands on the safe side	GDC 23 · SSR-2/1 Req.26
Interlock	Act only under defined conditions, or act without fail once they are met	GDC 21·24

The clearest fail-safe there is: the control rod.

Cut the power and the electromagnet lets go, so the rod falls **under gravity**. The failure called “loss of power” *is* the safe outcome called “shutdown”.

Had the design needed electricity to push the rod in, a blackout would mean **no shutdown at all**.

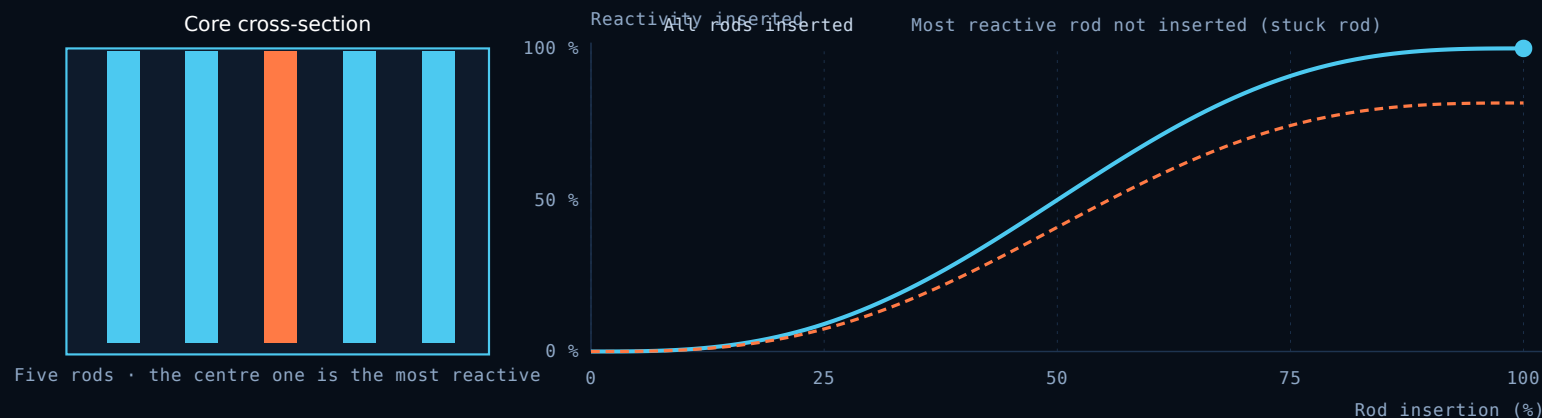
Same part, opposite logic

06 Shutdown systems

Safety function ①: how to kill reactivity for certain. Why control rods and boron injection are both needed, and what calculation fixes the performance each must deliver.

The first and last part of the stroke are **inefficient** - which is why drop time becomes a requirement

- Reactivity goes in fastest while the rod crosses the **high-flux centre** of the core - that is why the curve is S-shaped
- **GDC 26** - the requirement must still be met assuming the single most reactive rod **fails to insert**



TIME SINCE RELEASE

3.0 s

REACTIVITY INSERTED (OF TOTAL)

100 %

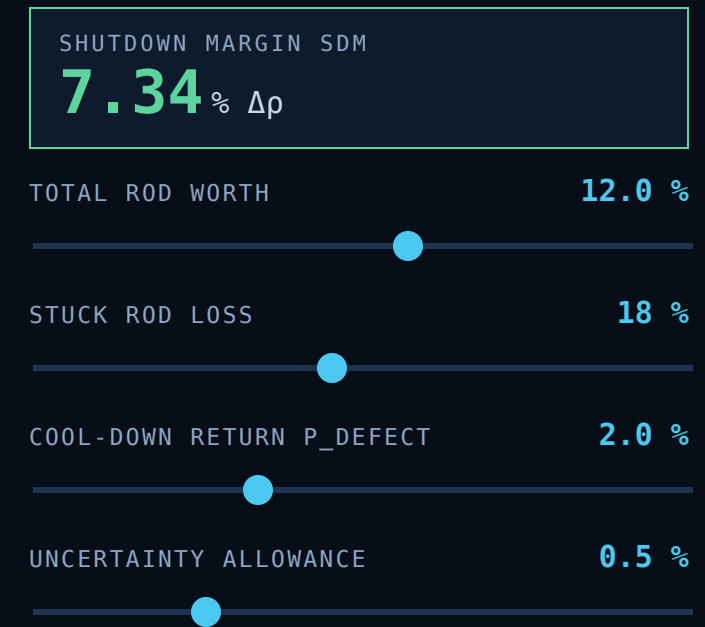
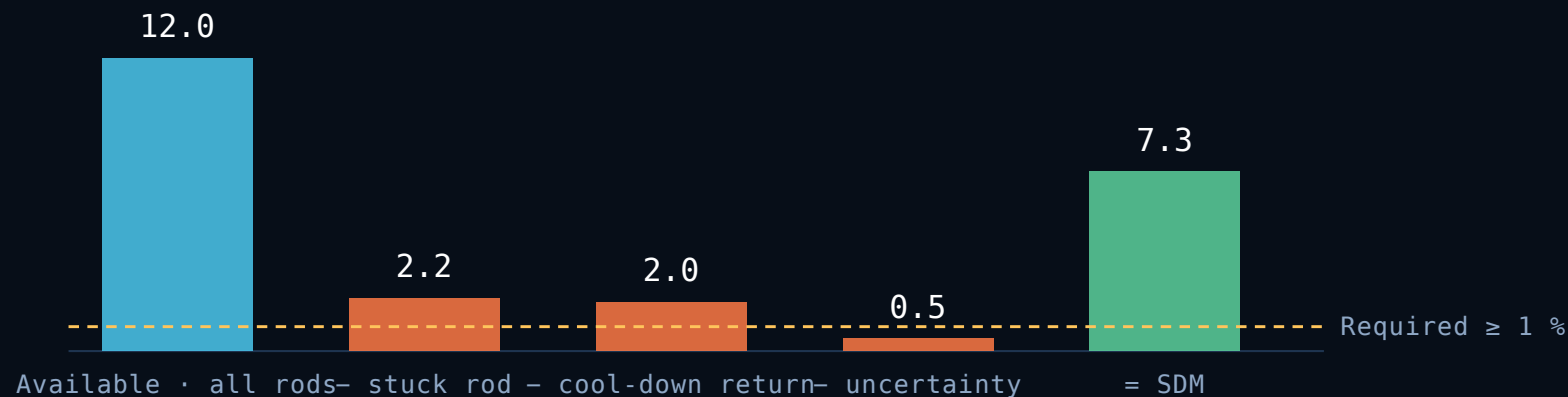
Reactor trip - replay

Assume a stuck rod

Cut the power, the magnet lets go and the rod falls under gravity - GDC 23 fail-safe.

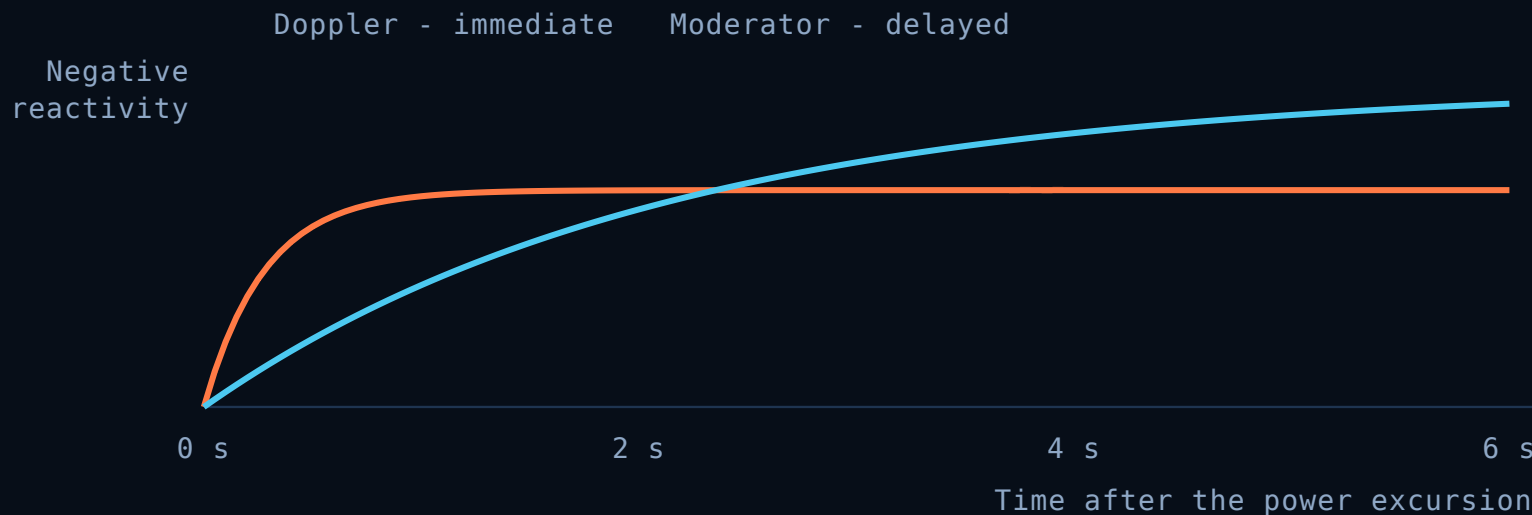
Shutdown margin - what you have minus what gets taken away

- As the plant cools the moderator gets denser and reactivity **comes back positive**. That much has to be covered in advance
- Move the sliders to see how each term eats into the margin - the defaults are the conditions of Exercise 6.2 in the text



What acts **before** any signal - the physics of the core itself

- **Doppler coefficient** -2 to -4 pcm/K - the fuel temperature responds at once, **immediate**. The first line against a rapid reactivity insertion
- **Moderator temperature coefficient** -10 to -60 pcm/K - acts only after the heat reaches the coolant, **delayed**
- With boron dissolved, a temperature rise **drives absorber out along with the water** and eats into that negative value → the case for a boron-free design



What a boron-free design gains

The whole boron system - storage, concentration control, purification, piping - goes, and the boron dilution accident goes with it. The moderator coefficient stays firmly negative all cycle.

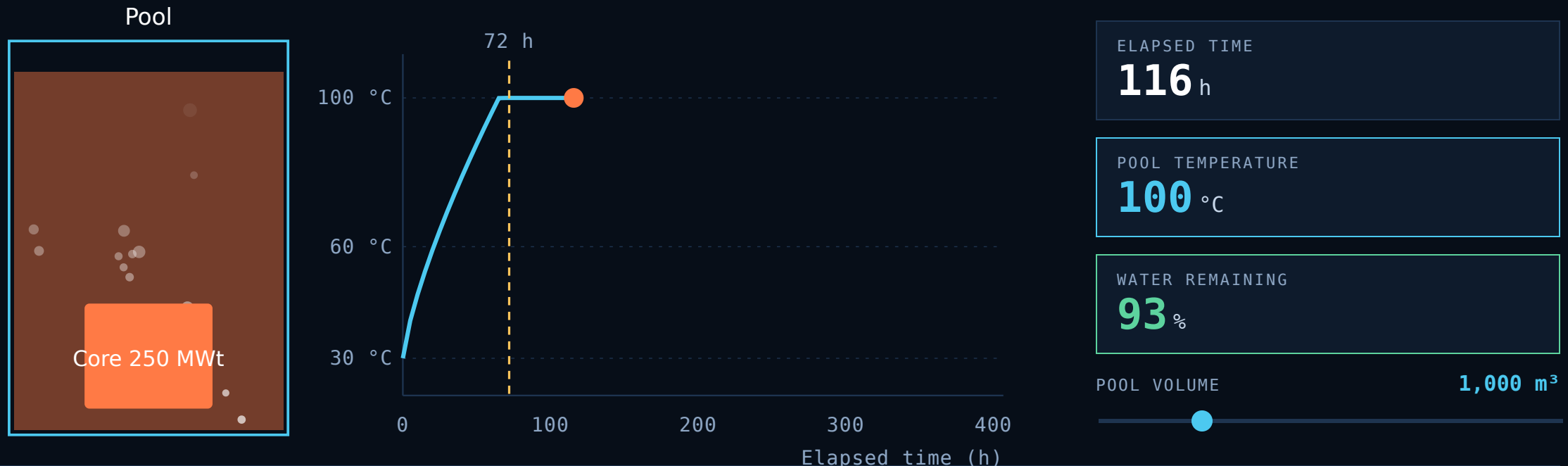
What it takes on: what satisfies the second independent system GDC 26 demands, and how to absorb a whole cycle of reactivity on rods and feedback alone.

07 Residual heat removal

Safety function ②: where the heat goes after shutdown. First pin down how much heat, for how long and to where, in numbers; then compare three ways of doing it.

“72 hours without operator action” is not magic - it is the calculation that sizes the tank

- The decay energy accumulated in 72 hours is about 315 GJ - taking it on sensible heat alone needs **about 1,000 t** of water
- Latent heat is **about eight times** the sensible heat, so it holds out far longer once boiling starts - at the price of a new design problem: venting the steam and making up the water

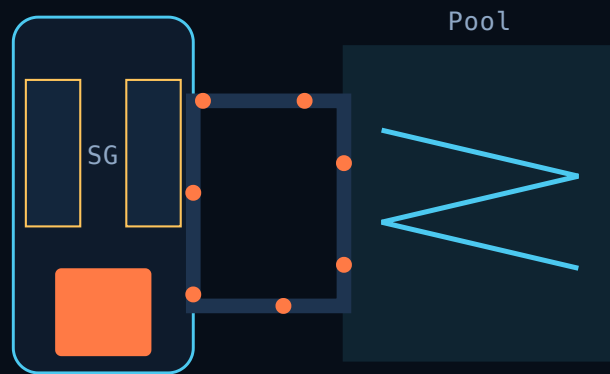


Calculation Way-Wigner decay heat integrated (250 MWt , $t_0 = 10^8 \text{ s}$) $\cdot c_p = 4.18 \text{ kJ}/(\text{kg}\cdot\text{K})$, $h_{fg} = 2,257 \text{ kJ/kg}$, initial water temperature 30 °C . A bounding estimate that sends all the heat into the pool - in reality heat exchanger capacity, stratification and heat loss all intervene

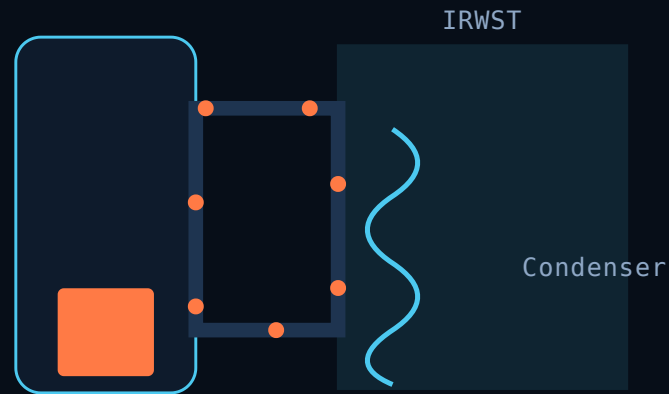
Where do you move the heat - **three** answers

- Only (a) **keeps the primary pressure boundary closed**. In exchange, it is useless if the steam generator itself has failed

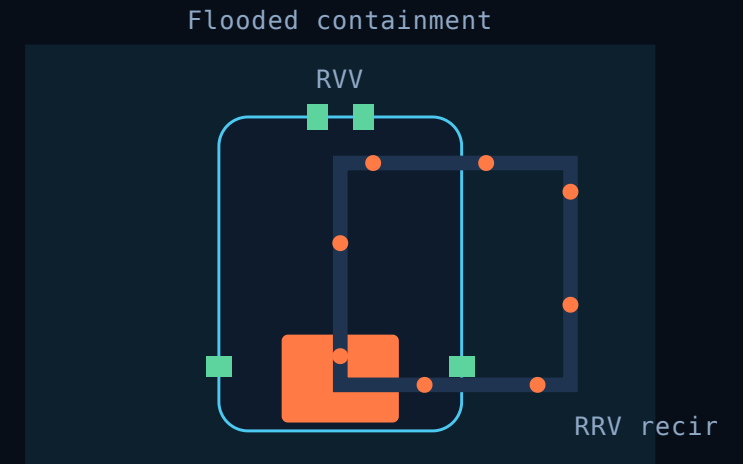
(a) Steam generator + pool heat exchanger (b) Passive condenser, submerged in the IRWS (c) Direct cooling in a flooded containment



IRIS EHRS · SMART PRHRS · NuScale DHRS



AP1000 PRHR HX · BWRX-300 ICS



NuScale ECCS (RVV and RRV open)

“72 hours” means different things in different designs

- ① time held without operator action · ② time held without external water · ③ time held without AC power
- The three definitions give very different numbers - when comparing, **always check which one is meant**

DESIGN	SYSTEM	PRINCIPLE	GRACE PERIOD, NO ACTION
SMART100	PRHRS · 4 trains	Secondary side plus an emergency cooldown tank	Safe shutdown within 36 h plus 36 h held = 72 h
i-SMR	PAFS / PCCS	Passive auxiliary feedwater for non-LOCA, passive containment cooling for LOCA	72 h without power, then air cooling
BWRX-300	ICS · 3 loops (about 33 MW each)	Reactor steam to a condenser in an atmospheric pool	7 days minimum
AP1000	PRHR HX + PCCS	A C-shaped heat exchanger submerged in the IRWST plus external shell spray	PCCWST drains over 72 h
CAREM-25	PRHRS · 2 trains	Horizontal U-tube condenser	36 h on one of the two trains
NuScale NPM	DHRS · 2 trains (100 % each)	A two-phase natural circulation closed loop with the condenser submerged in the pool	Long-term heat removal into the reactor pool

Source K.H. Bae et al., *Nucl. Eng. Des.* 379 (2021) 111247 (SMART100) · IAEA SMR Booklet 2022 (i-SMR, BWRX-300, CAREM, NuScale) · NRC ADAMS ML15230A043 (AP1000). All are vendor-stated values

08 Safety injection

The second half of safety function ②: how to put back the coolant that was lost, and the five quantitative acceptance criteria the regulator imposes.

The **pass mark** for a licensing safety analysis - learn these exactly

CLAUSE	CRITERION	VALUE
(b) (1)	Peak cladding temperature (PCT)	Shall not exceed 2,200 °F (about 1,204 °C)
(b) (2)	Maximum local oxidation	Nowhere more than 0.17 times the pre-oxidation cladding thickness
(b) (3)	Total hydrogen generation	No more than 0.01 times the hypothetical amount if all the cladding reacted
(b) (4)	Coolable geometry	Core geometry changes shall leave it coolable
(b) (5)	Long-term cooling	Decay heat shall be removed for an extended period after the initial ECCS success

WHY TEMPERATURE IS THE CRITERION

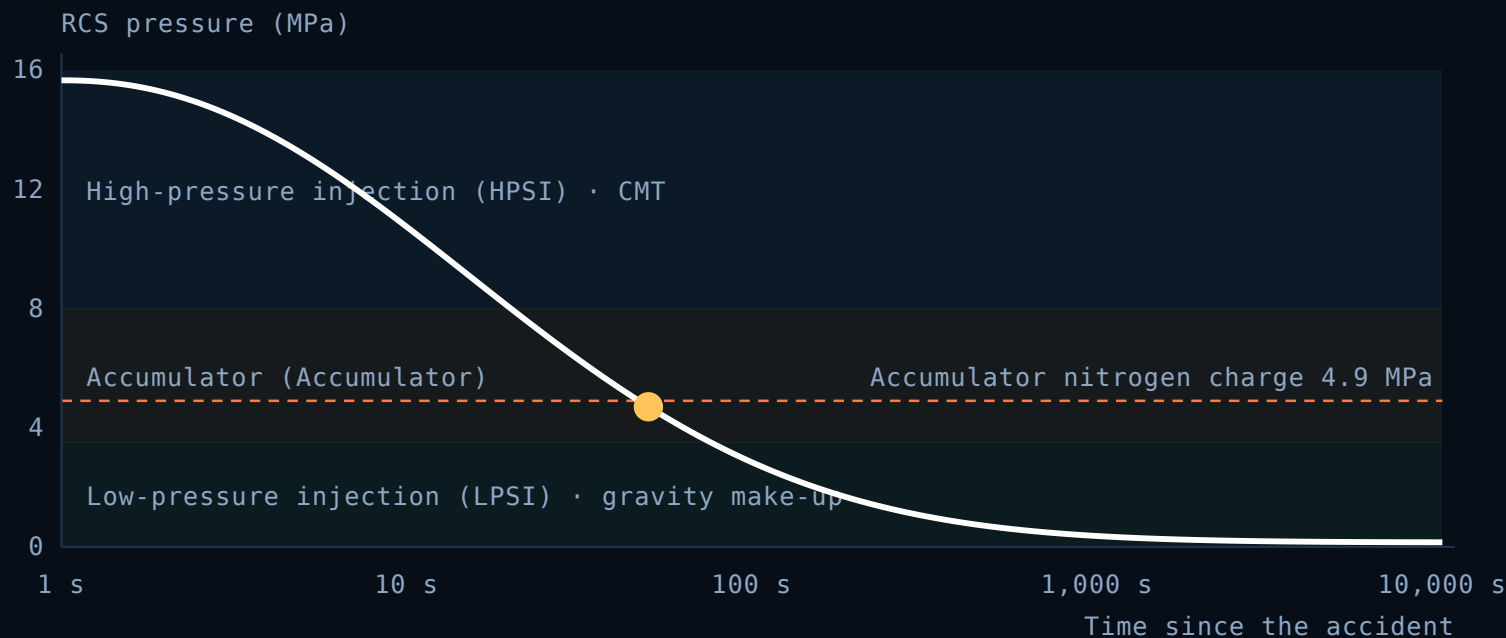


The reaction is exothermic. The hotter it gets the faster it runs, and its own heat raises the temperature further - **positive feedback**.

1,204 °C is the line below which that runaway does not start and the cladding keeps enough ductility not to shatter on quench. The hydrogen it produces becomes an explosion hazard inside containment, so (b)(3) caps the total - **this is where Chapters 8 and 9 join up**.

As the pressure falls they **hand over in turn** - no single system can do it

- Early on the pressure is too high for a low-pressure pump to push against; later the high-pressure system cannot deliver enough flow
- So **each pressure band gets its own means of injection** - that is the skeleton of an ECCS



RCS PRESSURE

4.7 MPa

SYSTEM NOW IN CHARGE

Accumulator

TIME SINCE THE ACCIDENT

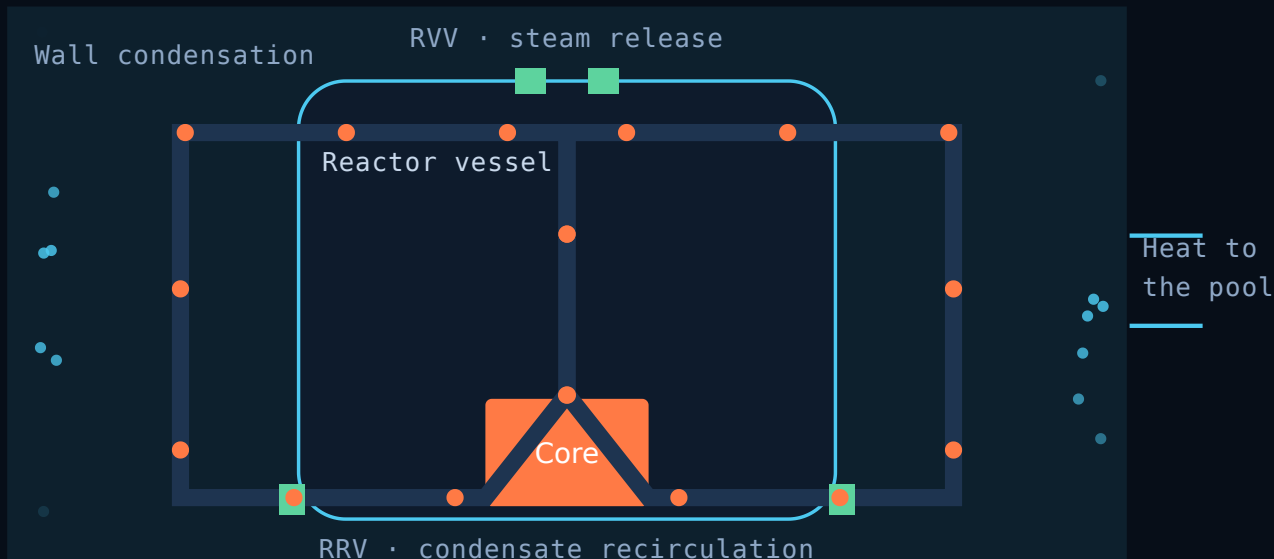
50 s

The accumulator opens on a check valve alone, with no valve operation - Category C passive. The CMT flows by gravity even at full RCS pressure.

Instead of adding water from outside, it keeps the water in and circulates it

- ① open the **RVVs** and vent steam into containment → ② it **condenses** on the submerged containment wall and the heat goes to the pool → ③ the **RRVs** return the condensate to the vessel
- Circulation holds with no water source outside the system, which is how **50.46(b)(5) long-term cooling** is met

Submerged containment vessel (CNV) · under vacuum in normal operation



EXTERNAL WATER SOURCE

Not needed

DRIVING FORCE

Condensation Δp

In an integral SMR the **safety injection system and the containment system are effectively one system**. We teach them apart, but in a real design the boundary is gone.

The i-SMR, with three EDVs and two ERVs, follows the same logic

09 Containment

Safety function ③, the last of the four barriers. Only one property is demanded of it - do not leak. And to hold that property, the heat has to go somewhere.

A containment necessarily has holes in it - do they close when it matters

- Pipes, electrical penetrations, personnel airlocks. If these do not close, no thickness of wall helps
- A plain **check valve** does not count as an automatic outboard isolation valve - it does not close in the direction you need to block

ARRANGEMENT	INBOARD VALVE	OUTBOARD VALVE
①	Locked closed	Locked closed
②	Automatic isolation	Locked closed
③	Locked closed	Automatic isolation
④	Automatic isolation	Automatic isolation

GDC 55
Lines penetrating the reactor coolant pressure boundary

GDC 56
Lines connected directly to the containment atmosphere - the outboard valve goes close to the containment

GDC 57
Other closed-loop penetrating lines - at least one outboard valve that is automatic, locked closed or remotely operated

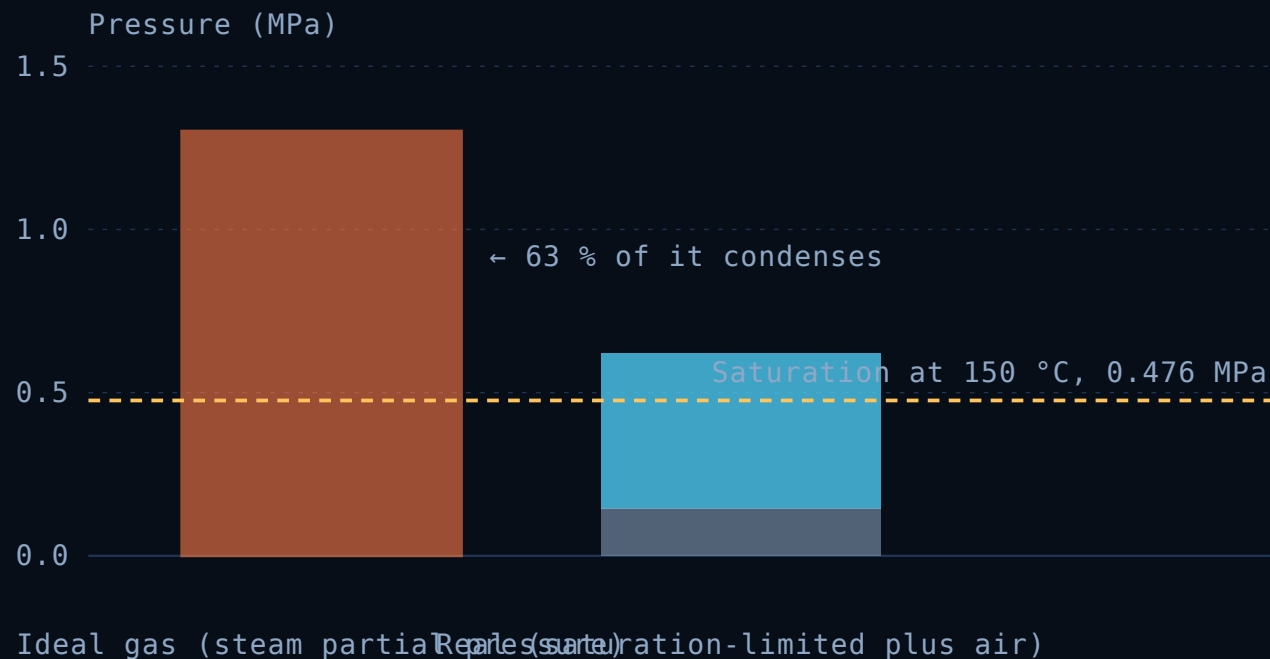
Source 10 CFR 50 Appendix A, GDC 16 and 50-57. GDC 52 requires that an integrated leak rate test be possible at design pressure - "it does not leak" has to be proved by measurement, not assertion

THINK OF IT AS HANYANG

Arrangement ④ is only for lines that need no flow in either normal or accident conditions - test, sample and drain lines.

Treat it as an ideal gas and you get it **wrong** - the excess **condenses**

- A computed steam partial pressure above the **saturation pressure** at that temperature is physically impossible → part of the steam turns liquid
- In the end **containment pressure is set by the equilibrium temperature, not by the mass of steam**. Which is why containment analysis is really a heat balance



IDEAL GAS STEAM PRESSURE

1.30 MPa

REAL CONTAINMENT PRESSURE

0.62 MPa

STEAM RELEASED

20.0 t

The air partial pressure is corrected for temperature only: $0.1013 \times 423/293 = 0.146$ MPa.

Worked example free volume 3,000 m³ · initial air at 1 atm and 20 °C · equilibrium temperature 150 °C · $R_v = 461.5$ J/(kg·K) · saturation pressure at 150 °C is 0.476 MPa. The point of the example is that the ideal gas approximation alone gives the wrong answer

There are only **three** places to put it - the atmosphere, water, or the wall

TYPE	WHERE THE ENERGY IS ABSORBED	ACTIVE / PASSIVE	REPRESENTATIVE DESIGN
(a) Large dry concrete plus spray	The atmosphere of a large free volume plus the sprayed water	Spray pump = active	Large LWRs · CAREM-25 (pressure-suppression, design pressure 0.5 MPa)
(b) Small submerged steel	Condensation on the steel wall → the pool outside	Passive · vacuum in normal operation	NuScale CNV - 4.5 m outer diameter by 23.1 m high
(c) Large steel, passively cooled	Evaporation on the outer shell plus natural-convection air flow	Passive · PCCWST drains over 72 h	AP1000 - once the water is gone, air cooling alone

Cut the free volume and the pressure rises **faster** - but the condensation heat flux on the wall rises too. Where the two balance is decided by the **condensation heat transfer coefficient**.

And a small amount of **non-condensable gas** is enough to collapse that coefficient. That is why NuScale holds its containment under **vacuum** - a variable no passive containment cooling analysis can ignore.

10 From requirement to system

Nine chapters pulled into a single traceability table. Read that table backwards and it becomes a design review checklist.

Design starts at the requirement and ends at the system - **read it the other way and you lose the reason**

Requirement	Safety function	System (ESF)
GDC 25 · 26 · 27	Reactivity control	Control rods · boron injection · (boron-free design)
GDC 34	Core residual heat removal	PRHR · DHRS · ICS · PAFS
GDC 35 · 10 CFR 50.46	Core coolant make-up	Accumulator · CMT · SI pump · RVV/RRV
GDC 16 · 38 · 41 · 50-57	Confining radioactivity	Containment · spray · isolation valves · PAR

Read this table backwards and it becomes a design review checklist

The same job under different names - set side by side

DESIGN	SHUTDOWN	RESIDUAL HEAT REMOVAL	SAFETY INJECTION	CONTAINMENT
NuScale NPM	Rods + boron	DHRS, 2 trains (100 % each) · two-phase natural circulation	No separate water source · 3 RVV + 2 RRV recirculation	Submerged steel CNV · vacuum
i-SMR	Rods (boron-free)	Non-LOCA: PAFS passive auxiliary feedwater	LOCA: 3 EDV + 2 ERV	Small steel CV + PCCS
SMART100	Rods + boron	PRHRS, 4 trains (secondary side + emergency cooldown tank)	4 CMT + 4 SIT (passive)	CPRSS · IRWST · PAR
BWRX-300	Rods + boron	ICS, 3 loops (about 33 MW each) · 7 days minimum	Integral isolation valves eliminate the large-pipe break	Below-grade PCV · about 16 m × 44 m
AP1000 (reference)	Rods + boron	One PRHR HX, submerged in the IRWST	2 CMT + 2 accumulators + IRWST · 4-stage ADS	Steel containment + PCCS
CAREM-25	FSS fast shutdown rods + SSS high-pressure boron tank	PRHRS, 2 trains · 36 h	2 MPIS accumulators, initiated by a rupture disc	Pressure suppression · design pressure 0.5 MPa
IRIS (historical)	Rods + 2 emergency boron tanks	EHRS, 4 passive trains · U-tubes inside the RWST	EBT + direct vessel injection (DVI)	Spherical steel · 25 m diameter · design pressure 1.3 MPa

Handed a design, can you fill in these four lines

① REACTIVITY CONTROL

What are the two independent means of killing reactivity? Is the shutdown margin still there with a stuck rod assumed?

② HEAT REMOVAL

What is the ultimate heat sink for decay heat after shutdown? How many days without action, and from what calculation does that number come?

③ COOLANT MAKE-UP

On losing coolant, what injects in each pressure band? How is long-term cooling established?

④ CONTAINMENT

Where does the containment absorb the released energy? What happens to the hydrogen?

NEXT · WEEKS 5 - 8

The code does not give you the answer

A code only computes the assumptions you feed it.

How you split the nodalisation, which correlation you used and over what range,
how much margin you left on the decay heat curve - those choices make the answer.

This material carries the requirements and the equations together so that every one of those choices can be justified.